



Trajectory optimization of train cooperative energy-saving operation using a safe deep reinforcement learning approach

Wenguang Niu¹ · Yonghua Zhou¹ · Xiangmeng Jiao¹ · Hamido Fujita^{2,3,4,5} · Hanan Aljuaid²

Accepted: 30 March 2025 / Published online: 11 April 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Energy-efficient optimization of train speed profiles can effectively reduce the traction energy consumption of urban rail transit systems. Existing reinforcement learning (RL) optimization models for optimizing train operation profiles do not proactively handle the utilization constraints of regenerative braking energy (RBE). For this reason, this paper proposes an optimization model of train energy-saving profiles under multi-train cooperative operations. A novel safe deep reinforcement learning algorithm, guided by heuristic rules, is developed to optimize energy-saving train driving strategies in various scenarios. To ensure safety during the agent's learning processes, a two-layer protection mechanism with soft constraint and truncation penalties is employed. Dynamic energy constraints are also introduced to enable the RBE utilization between trains. The simulation experiments using a real metro line data show that the proposed model and algorithm not only generate safe and energy-efficient profiles that meet metro operational constraints but also maximize the RBE utilization between trains, significantly reducing traction energy consumption.

Keywords Safe deep reinforcement learning · Reward design · Energy saving · Dynamic energy constraints

1 Introduction

Urban rail transit (URT) has achieved rapid development due to its advantages of high speed, convenience, punctuality, and safety [1]. However, the continuous expansion of the operation scale of the railroad network has also brought about a huge consumption of electric energy. The energy consumption of train operation accounts for nearly 50% of the total energy consumption of the URT systems [2], so it is of great practical significance to take reasonable measures to reduce

the energy consumption of train operation and thus to reduce the cost of URT operation.

The current research on reducing the energy consumption of train operations can be mainly categorized into two approaches. One class of methods is to optimize the train's station-to-station driving strategies, such as solving energy-saving train operation profiles given line parameters, speed limits and running times with heuristic search algorithms [3] and RL methods [4]. Another class of methods considers RBE between multiple trains, using heuristic algorithms to improve the utilization of regenerative braking energy (RBE) by maximizing the traction and braking overlap time of trains at the same substation [5, 6].

Under the safety protection of the automatic train protection (ATP) system, the automatic train operation (ATO) system can dynamically adjust train operation strategies by tracking the pre-given target speed profiles, thus realizing the automatic train operation control [7]. Designing safe, smooth, punctual, and energy-efficient train operating profiles is crucial for high-performance ATO. As an important branch of artificial intelligence (AI), RL leverages its strong exploration and adaptive learning capabilities to autonomously optimize optimal strategies through interaction with the environment. Optimizing these profiles is a

✉ Yonghua Zhou
yhzhou@bjtu.edu.cn

¹ School of Automation and Intelligence, Beijing Jiaotong University, Beijing, China

² College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia

³ Faculty of Science, University of Hradec Kralove, Hradec Kralove, Czech Republic

⁴ Malaysia-Japan International Institute of Technology, University Teknologi Malaysia, Kuala Lumpur, Malaysia

⁵ Regional Research Center, Iwate Prefectural University, Takizawa, Japan

complex multi-objective problem, as varying railroad conditions and speed limits make it difficult to establish a generalized reward function. Additionally, random exploration during early agent learning processes may breach train safety constraints, so learning safety to satisfy constraints should be guaranteed throughout the learning processes. Efficient RBE utilization is crucial to energy saving, but current RL-based train profile optimization strategies do not proactively consider RBE constraints. To address this issue, this study proposes a safe deep reinforcement learning (SDRL)-based train-driving strategy optimization method, which proactively considers the synergistic utilization of RBE between trains. The main contributions of this study are described as follows:

- (1) A time-step-based optimization model of energy-saving profiles is proposed under multi-train coordination, simultaneously considering train energy-saving profile optimization and RBE utilization.
- (2) The concept of dynamic energy constraints (DECs) is proposed, and an instantaneous energy-based RBE utilization model is established on this basis. Incorporating DEC into the learning process of an agent can adjust the traction time of the train, help the train absorb the available RBE, and reduce the overall energy consumption of the train's operation.
- (3) Based on real subway line data, we designed a DDPG-DEC-based train speed profile optimization algorithm, which takes into account the actual constraints of train operations. To ensure the safety of training processes, a reward function containing a two-layer safety protection mechanism is designed to guide the agent to adaptively conduct safe learning for energy-saving profiles under different scenarios.

The rest of the paper is organized as follows. In Section 2, we review the related research work on train energy-saving optimization. In Section 3, we develop a time-step-based train kinematics model with a train RBE utilization model, which introduces the concept of DECs. The DRL model for train operation optimization is developed and the algorithm architecture of DDPG-DEC is designed in Section 4. Simulation experiments are demonstrated in Section 5. Section 6 makes a conclusion.

2 Literature review

Many scholars have devoted themselves to the study of energy-saving train trajectory optimization (ETTO) [8]. Howlett et al. [9] demonstrated the energy-saving operating conditions of a train on a smooth track, including

maximum acceleration (MA), cruise (CR), coasting (CO), and maximum braking (MB), based on Pontryagin's maximum principle (PMP). Liu et al. [10] further analyzed the optimal control sequence of the switching points of train operating conditions by using the PMP. Albrecht et al. [11] investigated the problem of the energy-efficient driving strategy of a train on a steeply sloping track and proved the existence and uniqueness of the optimal driving strategy. Many researchers also solved the ETTO problem by heuristic algorithms. Lechelle et al. [12] designed a genetic algorithm to solve the energy-saving speed profile and achieved 7% energy saving. Lu et al. [13] proposed a distance-based train trajectory search model and compared the performances of genetic algorithm and ant colony optimization (ACO) algorithm for solving the ETTO problem. Sicre et al. [14] proposed a fuzzy parametric genetic algorithm to solve the optimal driving strategy based on an accurate simulation of the high-speed train movement model.

Artificial intelligence technology has reinvigorated ETTO's research efforts. Yin et al. [4] used RL for the design of train operation profiles, which saved 10% energy consumption while ensuring punctuality and passenger comfort. With the rapid development of deep learning (DL) [15], DRL algorithms, derived from the combination of neural network (NN) [16] and RL, have been applied to solve EETO problems [17–21]. Zhou et al. [17] realized model-free continuous train motion control by fusing expert experience into two algorithms, deep deterministic policy gradient (DDPG) and normalized advantage functions (NAF). Shang et al. [18] built a reference system based on the DRL algorithm to deal with constraints during train transportation, and the energy-saving effect exceeds the traditional genetic algorithm. Zhang et al. [19] used an event-driven DDPG-based algorithm to solve the train trajectory and introduced a dynamic excitation system to ensure the safe operating intervals of adjacent trains. Lin et al. [20] designed a gated recurrent unit (GRU) based deep Q-Network (DQN) algorithm with a reward function designed to include on-time performance, energy consumption, and stopping position to achieve a multi-objective speed profile optimization for freight trains. Safe RL [21] incorporates safety constraints into the training framework by restricting the agent's exploration space and designing reward functions that penalize risky behaviors. This approach plays a crucial role in ensuring the safety of train operations. Wang et al. [22] proposed an energy-saving maximum entropy deep reinforcement learning algorithm that balances multiple optimization objectives and improves energy efficiency by about 20%. They used the speed limit as a soft constraint, and the agent would be negatively rewarded when a train showed speeding behaviors. Zhao et al. [23] proposed a Shield protection mechanism and designed a Shield SARSA algorithm to protect the learning

process of the speed profile of a high-speed train, and the experimental results show that the protection mechanism effectively reduced the unsafe actions of the agent during the learning process. Furthermore, the agent's learning process can be modeled as a Constrained Markov Decision Process (CMDP) [24] to regulate its risky behaviors. Solutions to CMDP problems include Constrained Policy Optimization (CPO) [25] and Constrained Update Projection (CUP) [26], among others. However, in multi-objective optimization tasks, such as train speed profile optimization, balancing the cost function and reward function design can be challenging.

The above studies have only considered single-train inter-station operation strategies, and in recent years, more and more scholars have focused on multi-train RBE utilization. For example, Sun et al. [27] treated dwell time as a decision variable and formulated a Mixed-Integer Linear Programming (MILP) model to maximize the utilization of RBE. Sahebi et al. [28] used genetic algorithms (GA) to adjust the train dwell time and headway time to improve the RBE utilization of trains. The integrated model proposed by Shang et al. [29] considers both single-train inter-station control and multi-train cooperation, with the inner layer using quadratic programming to optimize train operation profiles and the outer layer optimizing train schedules through a parallel particle swarm algorithm, which reduces the energy consumption of the entire metro line operation. In addition to optimizing train schedules, another effective way to increase RBE utilization is to adjust the speed profiles of trains running between stations, a method that maximizes RBE utilization. Liu et al. [30] proposed a cooperative control model for neighboring trains, in which the front train adopts an energy-efficient driving strategy, and the rear train absorbs the RBE generated by the front train through maximum acceleration when the front train brakes. Sun et al. [31] realized three-train cooperative RBE absorption by replacing the original partial train driving strategy by coasting-traction-coasting by adjusting the speed profile. Jin et al. [32] proposed a real-time train control method based on train-to-train communication, where an acceleration phase is inserted during the braking state of other trains to improve the utilization of RBE. Su et al. [33] proposed a multi-train cooperative control optimization method based on multi-agent reinforcement learning (MARL), which reduces the net energy consumption by more than 11.1% compared to the single-train optimization method.

In the current literature about multi-train cooperative operation, the time-step-based calculation method of RBE utilization is rarely addressed. In addition, the DRL algorithms for train cooperative energy-saving operations should proactively consider critical constraints, such as DEC, to avoid unintended explorations of an agent.

3 Problem Formulation

In this section, we establish a time-step-based train movement model. RBE utilization can be significantly enhanced by adjusting the train speed profile to increase overlap time within a specified schedule. Consequently, we extend the traditional traction-braking overlap time calculation model to include a non-braking-braking model. To accurately address the RBE utilization problem, we establish a time-step-based RBE utilization model and introduce the concept of DEC.

3.1 Train movement model

The model is designed for a time-step-based DRL algorithm, so Δt is used to denote the minimum control time interval. The running time variation of a train can be described as

$$t_{i+1} = t_i + \Delta t \quad (1)$$

where i denotes the index of time series, $0 \leq i \leq I - 1$, and I is the total number of time steps. The initial running time of the train $t_0 = 0$.

In general, the single-mass model is the most commonly used model for solving the optimization problem of train running profiles. The basic motion equation of a train can be expressed as

$$\frac{d^2 x_i}{dt^2} = \frac{F}{M} \quad (2)$$

$$\frac{dx_i}{dt} = v_i \quad (3)$$

where M is the mass of a train. x_i and v_i denote the position and velocity of a train at the moment corresponding to time step i , respectively. F denotes the combined external force on the train, expressed as

$$F = u(v_i) - f_r(v_i) - f_g(x_i) \quad (4)$$

$$f_r(v_i) = \frac{Mg}{1000} r(v_i) \quad (5)$$

where $u(v_i)$ denotes the traction or braking force exerted on the train at speed v_i . $f_r(v_i)$ denotes the basic operating resistance of a train, generally described by the Davis equation $r(v_i) = a + bv_i + cv_i^2$, where a , b and c are empirical constants. $f_g(x_i)$ denotes the external force generated by gravity when a train is on a gradient section, expressed as $f_g(x_i) = Mg \sin(\theta(x_i))$, where g is the acceleration of gravity, and $\theta(x_i)$ denotes the angle of the track's gradient.

When time interval Δt is used to control a train, the following recursive formulas can be used to iteratively solve the

speed and position of the train:

$$\begin{cases} v_0 = 0 \\ v_{i+1} = v_i + \frac{F}{M} \Delta t \end{cases} \quad (6)$$

$$\begin{cases} x_0 = 0 \\ x_{i+1} = x_i + v_i \Delta t + \frac{F}{2M} \Delta t^2. \end{cases} \quad (7)$$

In addition, the following constraints should be satisfied considering the actual constraints of train operation processes:

$$\begin{cases} v_0 = v_{T_a} = 0 \\ x_{T_a} - x_0 = D \\ 0 < v_i < v^{\lim}(x_i) \\ U^{\min} \leq u(v_i) \leq U^{\max} \end{cases} \quad (8)$$

where v_0 is the initial speed of a train. v_{T_a} is the speed of a train when it arrives at the destination and stops, where T_a indicates the actual running time of the train, which should be as close as possible to the planned running time T_p . x_0 and x_{T_a} denote the initial position and stopping position of a train, respectively. D indicates the distance between stations of current operating range. $v^{\lim}(x_i)$ indicates the speed limit value of a train in a section of the current line, to ensure the safety of train operations, such that the train speed is prohibited to exceed the speed limit. $U^{\min}$ and $U^{\max}$ indicate the lower limit of braking force and the upper limit of traction force, which depends on the characteristics of the train's traction and braking systems, as well as the speed of a train at the current moment.

The traction energy consumption of a train during operations from station n to the station $n+1$ can be expressed by

the following equation:

$$E_n^{\text{tr}} = \sum_{i=0}^{I-1} \max\{0, u(v_i)\} \cdot v_i \Delta t. \quad (9)$$

When there is no available RBE, minimizing E_n^{tr} is the goal of the train energy-saving curve optimization problem, also ensuring comfortable, safe and punctual train operations.

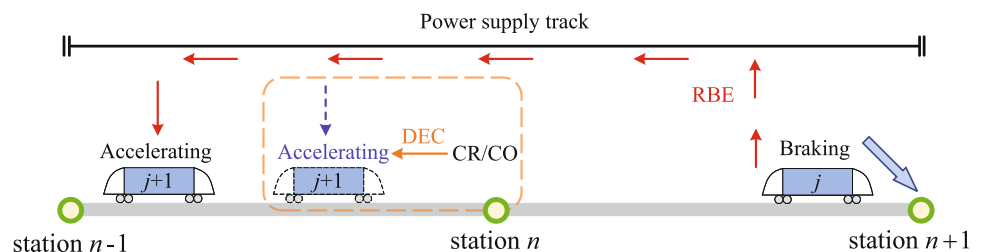
3.2 Utilization of BRE under dynamic energy constraints

3.2.1 Non-braking-braking overlap time calculation model

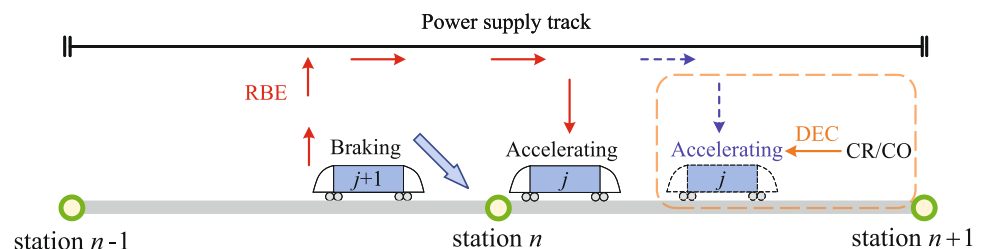
Figure 1 demonstrates the absorption process of RBE in neighboring trains under two scenarios, where the red arrows indicate the flow direction of RBE in the general case and the blue dashed arrows demonstrate the flow direction of RBE after adjusting the train operating speed profile.

Taking Fig. 1 (a) as an example, train j about to enter station $n+1$ is braking, and the RBE generated by braking is fed back to the contact network. At the same time, train $j+1$, which has just left station $n-1$, is accelerating, so the BRE generated by train j can be directly absorbed by train $j+1$, which can reduce the power consumption from a substation. Considering another scenario, when train $j+1$ is in the state shown in the dashed box in Fig. 1 (a), e.g., cruising or coasting, it cannot absorb the RBE generated by train j . If train $j+1$ adjusts its operation state to accelerate, it can absorb the RBE generated by train j .

Fig. 1 Direction of RBE flow during neighboring train operation



(a) Flow change of RBE after considering DEC in case 1



(b) Flow change of RBE after considering DEC in case 2

More specifically, the train operating profile adjustment process is shown in Fig. 2, where the two blue curves represent the original energy-saving operating speed profiles of trains j and $j+1$, respectively. Since the time of train j in the braking phase overlaps with the time of train $j+1$ in the cruising and coasting phases, the working condition of train $j+1$ in the overlap time range can be adjusted to traction to absorb the RBE. The adjusted speed curves are shown as red dashed lines in Fig. 2.

We use $(t_{j,n}^{tr}, t_{j,n}^{br}, t_{j,n}^{st})$ to describe the critical condition switching time of train j departing from station n . These three times represent the maximum traction start time, the maximum braking start time and the stopping time of train j , respectively. Therefore, we can use the following equation to describe the overlap time in the scenario of Fig. 1 (a):

$$t_1^{ol} = \begin{cases} 0 & t_{j+1,n-1}^{br} \leq t_{j,n}^{br} \\ \min\{t_{j,n}^{st}, t_{j+1,n-1}^{br}\} - \max\{t_{j+1,n-1}^{tr}, t_{j,n}^{br}\} & \text{otherwise} \\ 0 & t_{j,n}^{st} \leq t_{j+1,n-1}^{tr} \end{cases} \quad (10)$$

Similarly, when train $j+1$ about to enter station n is braking and train j departing from station n is in an accelerating, cruising, or coasting state, the overlap time equation in the scenario of Fig. 1 (b) can be derived as

$$t_2^{ol} = \begin{cases} 0 & t_{j,n}^{br} \leq t_{j+1,n-1}^{br} \\ \min\{t_{j+1,n-1}^{st}, t_{j,n}^{br}\} - \max\{t_{j,n}^{tr}, t_{j+1,n-1}^{br}\} & \text{otherwise} \\ 0 & t_{j+1,n-1}^{st} \leq t_{j,n}^{tr} \end{cases} \quad (11)$$

3.2.2 Dynamic energy constraint

During train operations, insufficient traction may prevent full absorption of available RBE, while applying maximum traction might increase energy consumption. To address this, we

introduce a DEC, which ensures the train to maximize the RBE absorption. Combined with the strong learning capabilities of DRL, this approach guarantees that the optimized driving strategy satisfies the original constraints.

Based on the overlap time calculated from (10) and (11), the RBE generated by the braking train during the overlap time range can be denoted as

$$E_{n'}^{rbe} = \sum_{i=k}^{k+K-1} \rho \sigma \min\{0, u(v_i)\} \cdot v_i \Delta t \quad (12)$$

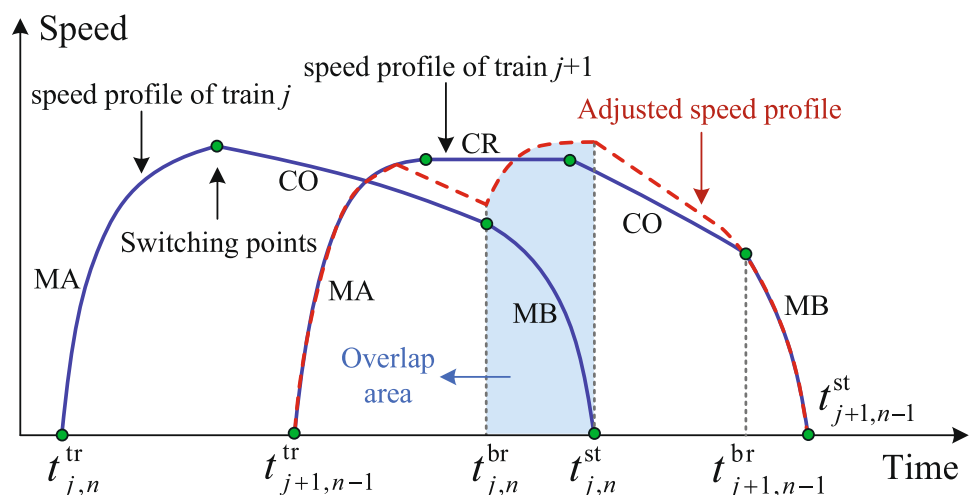
where K is the number of time steps of train control during the overlap time, and k is the index of the time sequence corresponding to the start of the overlap time. σ and ρ denote the regeneration coefficients of train braking energy and loss coefficients during RBE transmission, respectively. n' denotes that the train that generates the RBE may come from any station that is in the same power supply zone as the traction train.

When a train has available RBE during operation, the agent shall output an action as traction to make the train absorb the RBE. As the train's speed continuously changes over time during traction and braking, the instantaneous energy it generates or consumes per unit time also varies dynamically. To ensure that the train can fully absorb the RBE from the braking train, the traction energy of the traction train in each time step shall satisfy the following dynamic energy constraints:

$$E_{n,i}^{tr} \geq E_{n',i}^{rbe} \quad (13)$$

where $E_{n,i}^{tr}$ and $E_{n',i}^{rbe}$ denote the traction energy required by the traction train and the RBE generated by the braked train in the i -th time step, respectively. They are expressed by the

Fig. 2 Adjustment of train speed profile under case 1



following two equations:

$$E_{n,i}^{\text{tr}} = \max\{0, u(v_i^{\text{tr}})\} \cdot v_i^{\text{tr}} \Delta t \quad (14)$$

$$E_{n',i}^{\text{rbe}} = \rho \sigma \min\{0, u(v_i^{\text{b}})\} \cdot v_i^{\text{b}} \Delta t \quad (15)$$

where v_i^{tr} and v_i^{b} denote the speed of the traction train and the speed of the braking train at the current moment, respectively. Specifically, substituting (14) and (15) into (13), the following constraints should hold for the traction force of the train:

$$u(v_i^{\text{tr}}) \geq \frac{\rho \sigma \min\{0, u(v_i^{\text{b}})\} \cdot v_i^{\text{b}}}{v_i^{\text{tr}}}. \quad (16)$$

According to (16), as the speeds of the traction train v_i^{tr} and braking train v_i^{b} change during operation, the traction force of the traction train must be continuously adjusted to satisfy the DEC. In other words, DEC guides the agent to output an appropriate traction force, enabling the train to achieve complete absorption of RBE.

Finally, the net energy consumption of the train during its operation from station n to station $n+1$ can be expressed as:

$$E_n = \begin{cases} E_n^{\text{tr}} & \lambda_n^{\text{rbe}} = 0 \\ E_n^{\text{tr}} - E_{n'}^{\text{rbe}} & \lambda_n^{\text{rbe}} = 1 \end{cases} \quad (17)$$

where λ_n^{rbe} indicates whether there is available regenerative energy in the current operating interval. Reducing the net energy consumption E_n of train operation is the optimization goal of this study.

4 Methodology

Energy-efficient train control between stations necessitates a series of decisions at various points in time to address different road conditions, speed limits, and energy-saving goals. DRL can effectively optimize long-time-series decision-making in complex environments through real-time policy adjustments. In this section, we first introduce the proposed model for cooperative energy-saving strategy optimization for trains, outlining the basic elements of RL. We then detail the design method for a heuristic rule-guided safety reward function. Finally, we present the specific structure of the DDPG-DEC algorithm.

4.1 Model framework

To reduce the overall train operation energy consumption, it is essential to implement an energy-saving inter-station operation strategy while also considering the cooperative utilization of RBE between trains. This study proposes an optimization model for cooperative train operation strategies that accounts for RBE utilization and introduces DEC during agent-environment interactions to maximize RBE utilization. This model enables upper-level speed trajectory planning by coordinating RBE utilization among trains under a pre-defined timetable, eliminating the need for high real-time computing power. In practical operation, trains accurately track the preloaded speed profile via the ATO system to ensure local safety and a certain level of disturbance resistance.

The model structure is depicted in Fig. 3. It uses train basic data, subway line data, and operational constraints as inputs to create a train operation environment for agent interaction. The agent learns energy-saving driving strategies between stations through experiences gained from this interaction.

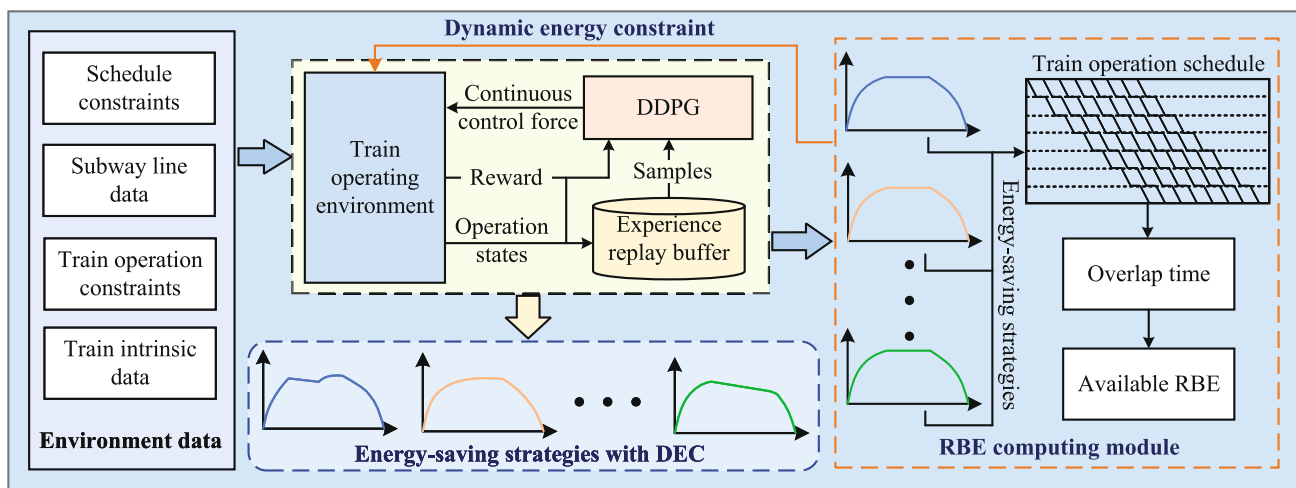


Fig. 3 Model framework for train cooperative operation strategy optimization

By integrating these strategies with the train timetable, the available RBE between trains can be calculated and incorporated as energy constraints into the agent's environment interaction. This updates the original energy-saving strategies to ones capable of absorbing RBE. Thus, by dynamically adjusting the operation strategy, DEC ensures maximum RBE absorption, effectively reducing the overall net energy consumption of train operation.

4.2 Reinforcement learning model construction

During train operation, the control force on a train depends solely on the current controller inputs and is independent of historical states. Thus, inter-station operation control can be modeled as a Markov decision process (MDP). DRL combines the decision-making mechanisms of RL with the perceptual capabilities of DL, allowing for efficient decision-making in continuous, high-dimensional, and large-scale state and action spaces. To optimize energy-saving strategies for ATO, the key elements of DRL need to be modeled, which include state, action, and reward.

4.2.1 State

In RL, the setting of the state will directly affect the learning efficiency of the agent. The state needs to contain enough information about the environment, such as the position and speed of a train, which is the basis for the agent to make control decisions about the train. In addition, to ensure that the agent accurately understands the environment and the task, we also select the acceleration and the current running time of the train as the environment state. The state space is defined as

$$\mathcal{S} = \{s_i = [t_i, v_i, x_i, a_i] \mid x_i \in [0, L], v_i \in [0, v_{\text{lim}}(x_i)], t_i \in [0, T_a]\} \quad (18)$$

4.2.2 Action

Train control systems output continuous train control forces, so we define the action of the agent as a continuous train control force coefficient:

$$\mathcal{A} = \{u_i \mid u_i \in [-1, 1]\} \quad (19)$$

when u_i is taken as 1 or -1, it means that the maximum traction force or the maximum braking force is applied to the train.

4.2.3 Reward

The reward signal serves as a guiding direction for an agent's learning, and a well-designed reward function helps balance

exploration and exploitation, preventing premature convergence to local optima. In DRL-based train energy-efficient operation control, challenges arise from safety constraint violations and sparse rewards. In traditional soft constraint penalty strategies, when a train exceeds speed limits or violates safety constraints, the agent receives a negative reward while the train continues to operate, which is infeasible in real-world scenarios. Moreover, key reward metrics such as energy consumption and travel time are only computed after the train completes its journey, leading to ineffective feedback and low learning efficiency due to reward sparsity. To enable the agent to adapt to diverse train operation environments and metro system requirements while balancing multiple optimization objectives, we propose a novel reward function design for generalizable train energy-efficient operation optimization.

First, a double-layer over-speed protection reward mechanism is established to ensure policy safety during the agent's learning process. As shown in Fig. 4 the green solid line indicates the minimum running time profile (MRTP) of a train, and the agent will not receive a speeding penalty when it explores the area below the MRTP (safe exploration area). The orange dashed line denotes the penalized speed threshold v_{sf} , and the red solid line denotes the truncated speed threshold v_{lim} . In the experiments in this paper v_{sf} is set to 90% of the speed limit value, while v_{lim} is the speed limit along the railroad. When the train speed exceeds v_{sf} but is less than v_{lim} , the agent receives a certain negative reward. If the agent's speed exceeds v_{lim} , the agent receives a larger negative reward for truncation and the episode ends. The mechanism helps the agent understand boundary conditions by allowing exploration near limits with soft constraint penalties while preventing dangerous behaviors through a hard constraint truncation strategy, which terminates training upon unsafe actions. This approach encourages the agent to respect safe boundaries by truncating negative rewards. Additionally, using expert train-driving knowledge, we decompose

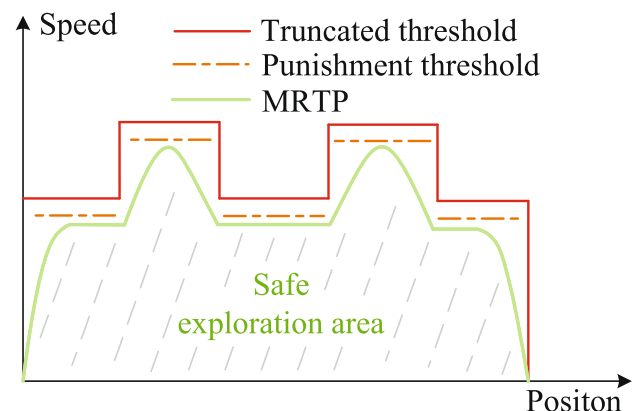


Fig. 4 Rewarding mechanism with double-layer over-speed protection

sparse rewards (e.g., energy consumption) to provide timely feedback during learning. Comfort is also considered, using impact rate as a heuristic rule to guide the agent. Overall, rewards are divided into three immediate categories—energy efficiency, comfort, and safety—along with sparse rewards at truncation and episode end.

- (1). Energy-saving reward r_e : We define as a positive number close to 0. When $|u_i| \leq \varepsilon$, it is approximated that the control force output by the agent is 0, at which time the train maintains the coasting state, while when $u_i < -\varepsilon$ or $u_i > \varepsilon$, it is assumed that the action output by the agent is braking or traction. First of all, to reduce the traction energy consumption of the train, the train should minimize the number of braking during operation, so unnecessary braking will give the agent a negative reward. In addition, G_{th} is defined as the threshold value for determining whether the current gradient g_i is a steep slope. When the train encounters a steep slope section, we want the train to attempt to tract on the uphill section to prevent the train speed from dropping too fast, while on the downhill section, we try to utilize the gravitational potential energy to glide in order to save energy consumption.

$$r_e = C_{e1} \begin{cases} u_i & \text{if } u_i < -\varepsilon \\ 0 & \text{else} \end{cases} + C_{e2} \begin{cases} u_i & \text{if } g_i > G_{th} \\ -|u_i| & \text{if } g_i < -G_{th} \end{cases} \quad (20)$$

- (2). Comfort reward r_c : In order to ensure the comfort of the passengers during train operation, we consider the impact rate generated by the train. A large change in the acceleration of the train in a short period of time will give a large negative reward to the agent.

$$r_c = -C_c \frac{|a_i - a_{i-1}|}{\Delta t} \quad (21)$$

- (3). Safety reward r_s : The maximum speed of the train during operation depends on the speed limit, and to ensure the safety of the train operation, the train's operating speed should be kept below the punishment threshold v_{sf} . In addition, when the train speed is small or even below the average speed v_{avg} , the train may not be able to reach the end of the route within the specified timeframe. Therefore, when the train runs beyond a reasonable speed range, the agent will be rewarded with some negative incentives. The safety reward r_s is described as

$$r_s = \begin{cases} -C_s & \text{if } v_i \geq v_{sf} \text{ or } v_i < v_{avg} \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

- (4). Truncated reward r_{trun} : When the state of a train exceeds the safety limits we set, e.g., the impact rate is greater than 1 or the speed exceeds the truncation threshold v_{lim} , the current training episode is truncated and the agent receives a large truncation penalty C_{trun} . The truncated reward r_{trun} formulates as

$$r_{trun} = \begin{cases} C_{trun} & \text{if } v_i > v_{lim} \text{ or } \frac{|a_i - a_{i-1}|}{\Delta t} > 1 \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

- (5). Episode ending reward r_{end} : If the train travels to the destination and stops at the exact location during that episode, the agent can receive an incentive positive reward C_{end} , to encourage the agent to control the train to reach the destination in the next episode. In general, the traction energy consumption of a train shows a negative correlation with the running time. Therefore, to ensure the energy saving and punctuality of a train at the same time, we also combine the weights of the train's running time error and traction energy consumption as part of r_{end} , which is described by

$$r_{end} = C_{end} + C_{d1} E^{tr} + C_{d2} |T_p - T_a| \quad (24)$$

Finally, the rewards are fused into a unified reward signal by assigning them with different weights to help the agent achieve multiple learning goals. The designed reward function is represented as

$$r = \underbrace{r_e + r_c + r_s}_{\text{Immediate reward}} + r_{trun} + r_{end} \quad (25)$$

where C_{e1} , C_{e2} , C_c , C_s , C_{trun} , C_{end} , C_{d1} , C_{d2} are adjustable weight hyperparameters. Based on the optimization objectives, engineering experience, and existing literature, we assign higher initial weights to safety and episode termination rewards to prioritize key objectives. Subsequently, lower weights are assigned to other rewards, with dynamic adjustments made according to the agent's learning performance to balance the impact of different rewards and enhance overall performance. It is worth noting that if the agent's action is not within the constraints, we set the default reward as 0 obtained by the agent.

4.3 DDPG-DEC strategy

Machine learning (ML) has become an effective solution in the real world [34], especially, DRL algorithms based on actor-critic architecture have been widely used in complex continuous control problems. DDPG [35] combines deep neural networks with deterministic policy gradients,

which can effectively solve high-dimensional, continuous state space and action space problems.

Figure 5 shows the interaction process with the environment and the policy update process of the designed DDPG-DEC algorithm. From Fig. 5, it can be seen that the driving strategy of a train does not only contain the control force of the agent's output u_i . To ensure the absorption of RBE by the train, the dynamic traction strategy u^{dec} generated based on DEC achieves the control of a train with higher priority when there is available RBE. Therefore, the energy-saving control strategy of the train can be expressed as

$$\pi(s_i) = \lambda_i^{\text{rbe}} u^{\text{dec}}(s_i) + (1 - \lambda_i^{\text{rbe}}) u_i. \quad (26)$$

It is worth noting that the control strategies generated by the DEC are only used for the update of the train operation state. And the sample data generated by the agent interacting with the environment will be stored in the experience replay buffer for the training of the main actor network and the main critic network, which have the parameters of θ^μ and θ^Q , respectively, and the corresponding target network parameters are $\theta^{\mu'}$ and $\theta^{Q'}$, respectively. According to the deterministic policy gradient theorem, the update formula for the main actor network can be obtained:

$$\nabla_{\theta^\mu} J \approx \frac{1}{B} \sum_b \nabla_u Q(s, u | \theta^Q)|_{s=s_b, u=\mu(s_b)} \nabla_{\theta^\mu} \mu(s | \theta^\mu)|_{s_b} \quad (27)$$

where B denotes the number of samples of a mini-batch from the experience replay buffer. The critic network is used for the approximation of the value function to update the parameters by minimizing the mean square error (MSE) between the Q value of the main network and the Q value of the target

Algorithm 1 DDPG-DEC for train operation control.

```

1: Randomly initialize critic network  $Q(s, a | \theta^Q)$  and actor  $\mu(s | \theta^\mu)$ 
   with weights  $\theta^Q$  and  $\theta^\mu$ 
2: Initialize target network  $Q'$  and  $\mu'$  with weights  $\theta^{Q'} \leftarrow \theta^Q, \theta^{\mu'} \leftarrow \theta^\mu$ 
3: Initialize replay buffer  $\mathcal{R}$ 
4: for  $episode = 1$  to  $N$  do
5:   for  $i = 1$  to  $I$  do
6:     if  $\lambda_i^{\text{rbe}} = 1$  then
7:       Select action  $u^{\text{dec}}(s_i)$  according to (16)
8:       Execute action  $u^{\text{dec}}(s_i)$  and observe new state  $s_{i+1}$ 
9:     else
10:      Select action  $u_i = \mu(s_i | \theta^\mu) + \mathcal{N}_i$  according to the current
      policy and exploration noise
11:      Execute action  $u_i$  and observe reward  $r_i$  and observe new
      state  $s_{i+1}$ 
12:      Store transition  $(s_i, u_i, r_i, s_{i+1})$  in  $\mathcal{R}$ 
13:      Sample a random minibatch of  $B$  transitions
       $(s_b, u_b, r_b, s_{b+1})$  from  $\mathcal{R}$ 
14:      Update the critic network by minimizing the loss function
      (28)
15:      Update the actor policy using the sampled policy gradient
      (27)
16:      Update the target networks according to (30)
17:    end if
18:  end for
19: end for

```

network. The loss function is represented as

$$L = \frac{1}{B} \sum_b (y_b - Q(s_b, u_b | \theta^Q))^2 \quad (28)$$

where y_b is the Q value of the target network, denoted by

$$y_b = r_b + (1 - \lambda_b) \gamma Q'(s_{b+1}, \mu'(s_{b+1} | \theta^{\mu'}) | \theta^{Q'}) \quad (29)$$

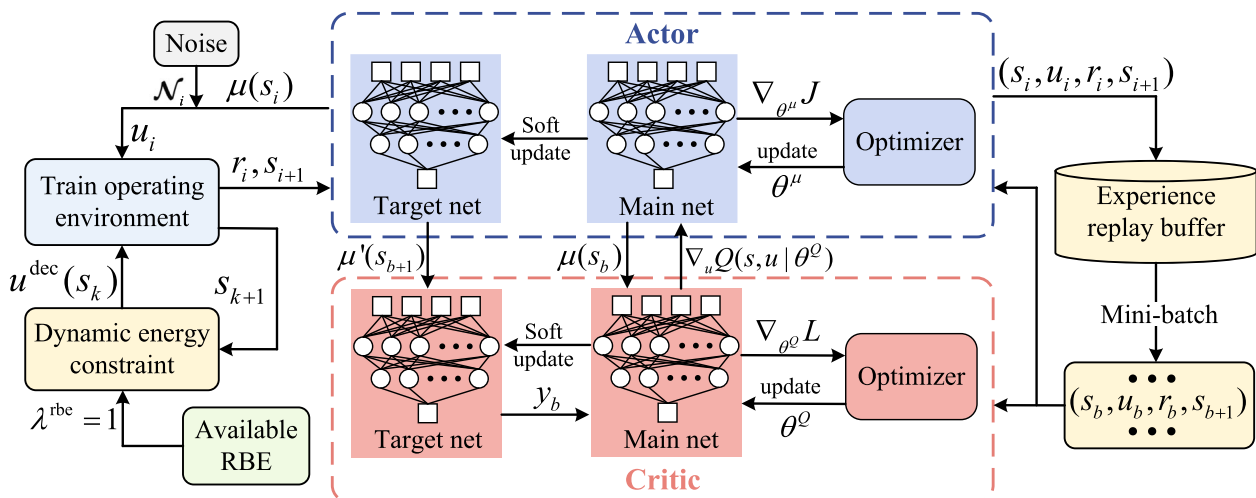


Fig. 5 Structure of DDPG-DEC

The default value of λ_b is 0, and λ_b takes the value of 1 only when s_b is the termination state.

To improve the stability of learning, the main network passes the parameters to the target network in a soft update manner:

$$\theta^{\mu'} \leftarrow \tau \theta^{\mu} + (1 - \tau) \theta^{\mu'}, \theta^{\mathcal{Q}'} \leftarrow \tau \theta^{\mathcal{Q}} + (1 - \tau) \theta^{\mathcal{Q}'} \quad (30)$$

where τ is a very small update parameter. The pseudo-code of the DDPG-DEC algorithm is shown in Algorithm 1.

5 Numerical experiments

This section details the test environment, simulation conditions, and numerical examples used to demonstrate the effectiveness of the proposed multi-train collaborative speed-profile optimization model for energy-saving in subway systems, as well as the performance of the proposed solution algorithm.

5.1 Simulation conditions

Simulation experiments were conducted using actual data [36] from the Yizhuang Line of the Beijing Subway (YLBS). The Yizhuang line is 22.7 kilometers long, including 14 stations and 6 power supply substations, and its inter-station distance and operation timetable are shown in Table 1. Train parameters include train mass $M = 1.94 \times 10^5$ Kg, train length $L_t = 110$ m, train maximum traction acceleration $a_{tr}^{\max} = 1$ m/s², maximum braking acceleration $a_{br}^{\max} = -1$ m/s². The characteristic curves of the train's traction force, braking force, and basic resistance are shown in Fig. 6.

Table 1 Inter-station spacing and driving plan of Yizhuang line

No.	Inter-station	Distance (m)	Running time (s)	Dwell time (s)
1	SJZ-XC	2631	190	30
2	XC-XHM	1274	108	30
3	XHM-JG	2366	157	30
4	JG-YZQ	1983	135	30
5	YZQ-WHY	992	90	35
6	WHY-WYJ	1539	114	30
7	WYJ-RJDJ	1280	103	30
8	RJDJ-RCDJ	1354	104	30
9	RCDJ-TJNL	2337	164	30
10	TJNL-JHL	2265	150	30
11	JHL-CQN	2086	140	30
12	CQN-CQ	1287	102	35
13	CQ-YZ	1334	115	45

The RBE transmission efficiency is generally dependent on the distance between trains, with efficiency decreasing as the distance increases. Since different transmission efficiency settings only affect the absolute magnitude of energy savings without influencing the algorithm's inherent performance, we assume in this paper that there is no transmission loss during RBE transmission for the sake of simplified calculations and result comparison. The substation configuration of the YLBS is shown in Fig. 7.

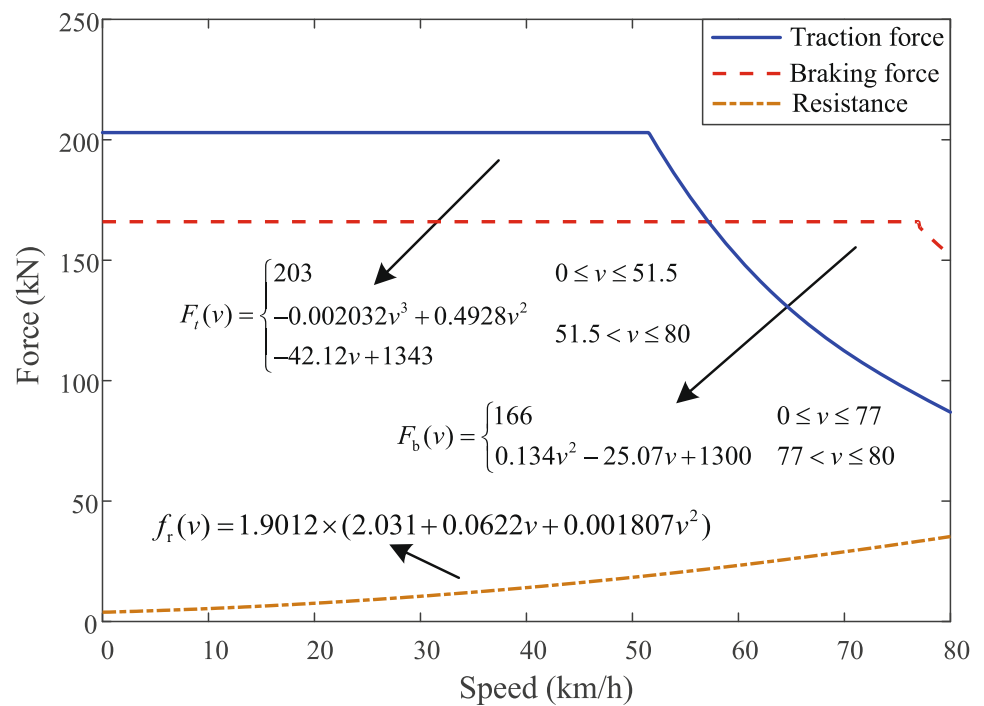
Considering the characteristics of train operations and actions in the simulation environment, along with the neural network's fitting capabilities [37], we use fully connected layers for both actor and critic networks. The actor network has 128 and 64 neurons in its two hidden layers, while the critic network has 128 neurons in both layers. Both networks use the ReLU activation function, with the actor network's output layer using a tanh activation to map agent actions to the train's control space. The replay buffer size is set to 100,000, and the mini-batch size is 128. Training uses the Adam optimizer with a learning rate of 0.0005 and a soft update factor τ of 0.005. Exploration noise follows a standard normal distribution with an initial exploration coefficient of 0.5, halving every 100 episodes. The control time interval Δt is set to 1 s to meet accuracy requirements in speed profile optimization, while during specific operations like braking, the train state update interval is reduced to 0.1 s for precise stop calculations. The simulation code was implemented in Python 3.9.18 with PyTorch 1.9.1 and executed on a computer equipped with an Intel Core i5-8300H @ 2.3 GHz, GTX 1060, and 16GB of RAM.

5.2 Experimental results

To test the protection effect of the designed SDRL algorithm, we selected SJZ-XC, the interval with the most complex speed limit on YLBS, as the test line. The iterative learning process is illustrated using 3D profiles of episodes, positions, and speeds alongside the speed limit plane in Fig. 8, with red curve segments indicating speeds that exceed the limit. In Fig. 8 (a), using only soft constraint penalties (the conventional approach) fails to fully ensure a safe learning process, leading to instances of train over-speeding. Increasing the penalty may compel the agent to reduce the train's traction force excessively to prevent over-speeding, resulting in excessively low speeds and failure to arrive on time. This reliance on soft constraint penalties complicates the design of the reward function. In contrast, Fig. 8 (b) demonstrates the iterative process guided by the proposed two-layer reward mechanism. Under this system, the agent avoids dangerous behaviors, does not exceed speed limits during iterations, and meets the required running time constraints.

In URT systems, multiple train timetables are often required to accommodate fluctuating passenger demand. The

Fig. 6 Train traction, braking, and basic resistance characteristic diagrams



originally planned running time for the SJZ-XC section is 190 s. To evaluate the adaptability of the proposed algorithm, we adjusted the running times to 180 s, 200 s, and 210 s for testing. As shown in Fig. 9, the train can proactively adjust to speed limits when significant changes occur. With shorter planned running times, the agent increases traction and adopts a faster acceleration strategy to meet time constraints. In the latter half of the interval, all speed profiles demonstrate consistent coasting strategies, indicating that the algorithm maintains a consistent optimization approach, ensuring smooth and efficient train control under varying conditions.

To assess the reliability of the proposed DDPG-DEC algorithm, energy-saving speed profiles were generated for all operational intervals of YLBS under the standard timetable, and corresponding performance metrics were calculated. As shown in Fig. 10 (a), the optimized position-speed curves remain within the speed limits, ensuring operational safety. In

intervals with complex speed limits or where single traction alone is insufficient to reach the station, the agent adaptively adjusts the traction output to maintain both on-time performance and safety.

The energy-saving optimization of the train's speed profile addresses multiple objectives, as illustrated in Fig. 10 (b), which shows the key metrics: energy consumption, punctuality error, and stopping position error. The results indicate a positive correlation between energy consumption and station distance in areas with gentler gradients, where longer station spacing requires more traction energy. However, in downhill sections like interval 3, despite nearly twice the station spacing compared to interval 2, less traction energy is consumed. This suggests that the agent efficiently uses gravitational acceleration to glide and conserve energy. Furthermore, the error between actual and planned running times is kept within 0.5 s, and stopping position errors remain within 0.2 meters.

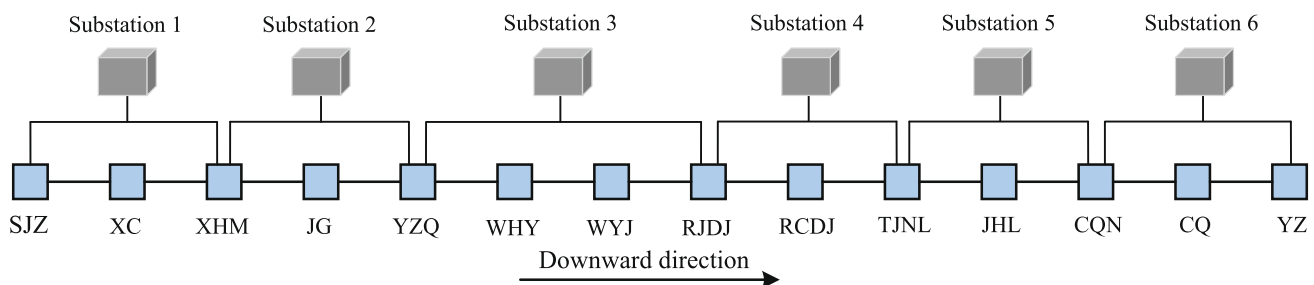
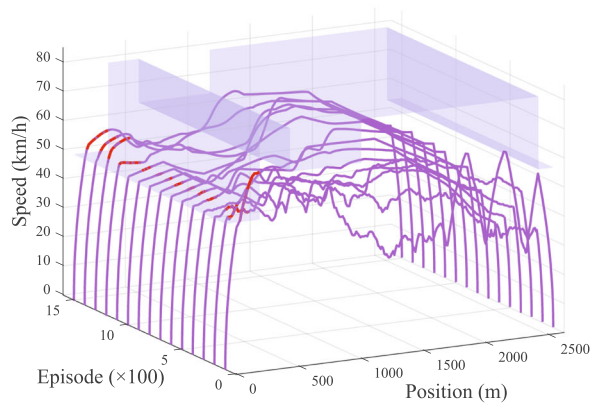
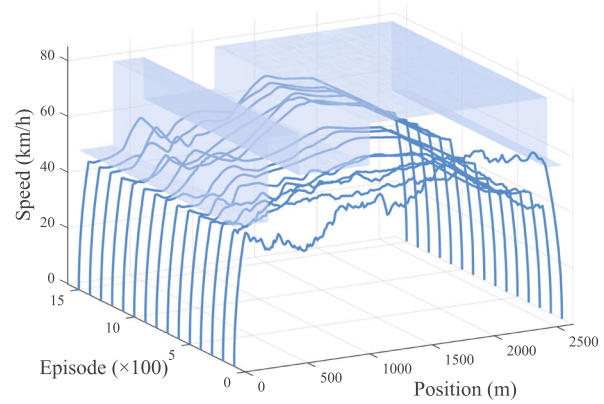


Fig. 7 Substation distribution of Yizhuang Line



(a) Learning processes guided by soft-constraint speeding penalties



(b) Learning processes guided by double layer of speeding protection

Fig. 8 Comparison of the learning process under two reward mechanisms

Once the train operation timetable is established, the designed DDPG-DEC algorithm can be applied to derive energy-efficient driving strategies for trains between stations, facilitating the calculation of RBE utilization. We investigate the feasibility of DEC for cooperative train control and its impact on energy-saving efficiency by examining two neighboring trains during operation. With a headway time of 260 s, the traditional traction-braking overlap method results in only 1.9 s of overlap time, leaving minimal RBE utilization. However, when trains operate under a cooperative control mode using the non-braking-braking approach, overlap time significantly increases. As long as trains are not under maximum braking, they can actively adjust traction to absorb RBE from neighboring trains, boosting the overlap time from 1.9 s to 70.8 s.

Figure 11 illustrates the driving strategy adjustments for two neighboring trains over five intervals with a 260 s headway. The blue dashed line represents the speed profile of

Train 1 (front train), and the green dotted line shows the speed profile of Train 2 (rear train). The red solid line indicates the adjusted speed profile after applying DEC. For instance, in Fig. 11 (c), when Train 2 is braking, the uncooptimized Train 1 is coasting and unable to utilize the available RBE. However, with DEC, the speed profile of Train 1 is adjusted (red line) to actively apply traction, fully absorbing the RBE. Despite DEC altering the traction time distribution, DRL ensures that the train still meets travel time constraints through its adaptive and exploratory capabilities. As shown in Fig. 11, the start and stop times remain nearly identical before and after the adjustment, demonstrating that the overall travel time remains unaffected.

We also analyzed the change in the traction energy of the trains to illustrate the energy-saving effect of Algorithm 1. Table 2 gives the original traction energy $E_{\text{org}}^{\text{tr}}$, the adjusted traction energy $E_{\text{adj}}^{\text{tr}}$, the adjusted net energy consumption $E_{\text{net}}^{\text{tr}}$, and the energy-saving efficiency of the trains in the five overlap intervals, where $\varphi = (1 - E_{\text{net}}^{\text{tr}}/E_{\text{org}}^{\text{tr}}) * 100\%$. It can be seen that the adjusted traction energy $E_{\text{adj}}^{\text{tr}}$ of the train under the DEC strategy is similar to that of the original traction energy $E_{\text{org}}^{\text{tr}}$, but due to the active absorption of RBE, the final net energy consumption $E_{\text{net}}^{\text{tr}}$ is reduced substantially. Based on the energy-efficient driving strategy adopted between each station, the application of Algorithm 1 further reduces the traction energy consumption by at least 10.94% and can save up to about 47.06% of the traction energy between other stations.

To verify the stability of the experimental results, we conducted multiple comparative experiments on DDPG-DEC, DDPG, and TD3 [38]. Taking the WYJ-RJDJ section as an example, Fig. 12 illustrates the learning process of average energy consumption and travel time. As shown in Fig. 12(a), compared to conventional DDPG and TD3, the introduction

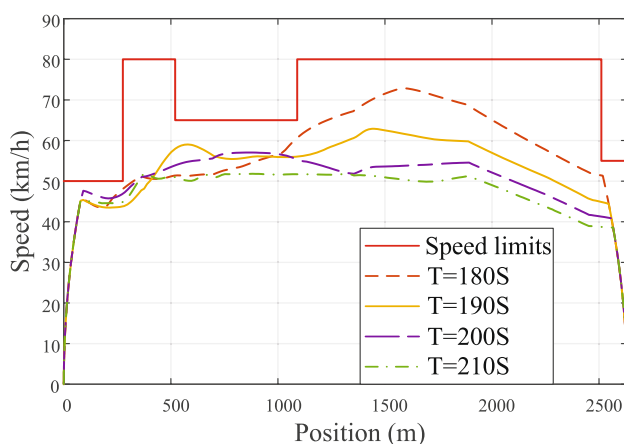
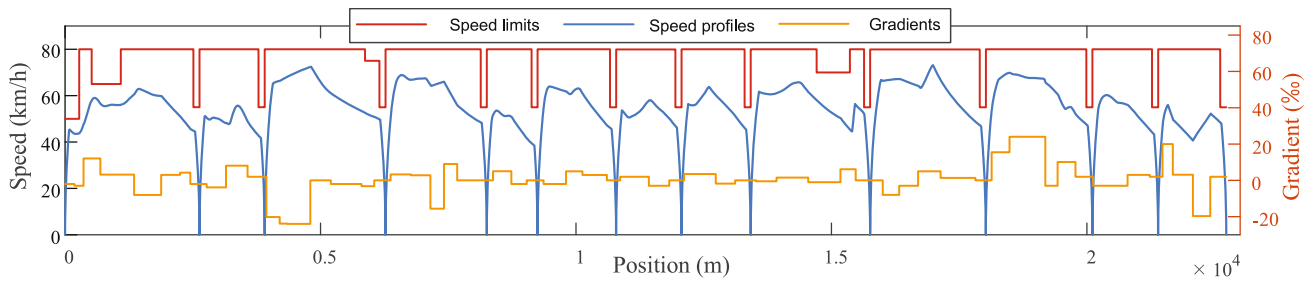
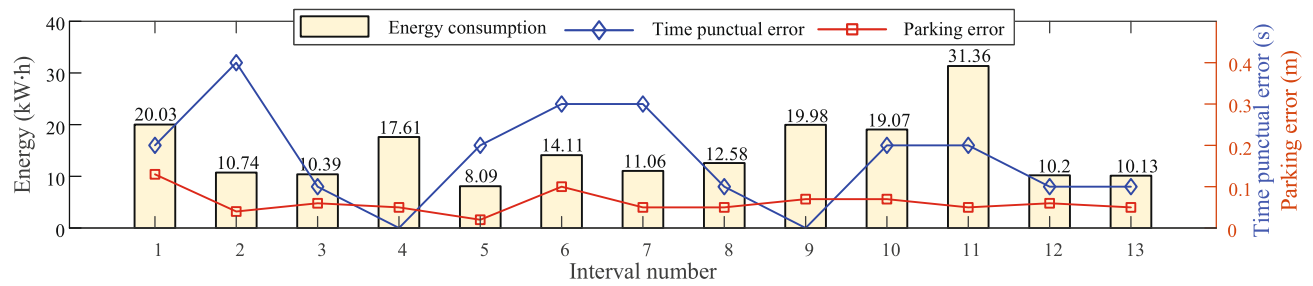


Fig. 9 Speed profile of SJZ-XC under different running time



(a) Speed profiles of all inter-stations of YLBS



(b) Evaluation indicators of all inter-stations of YLBS

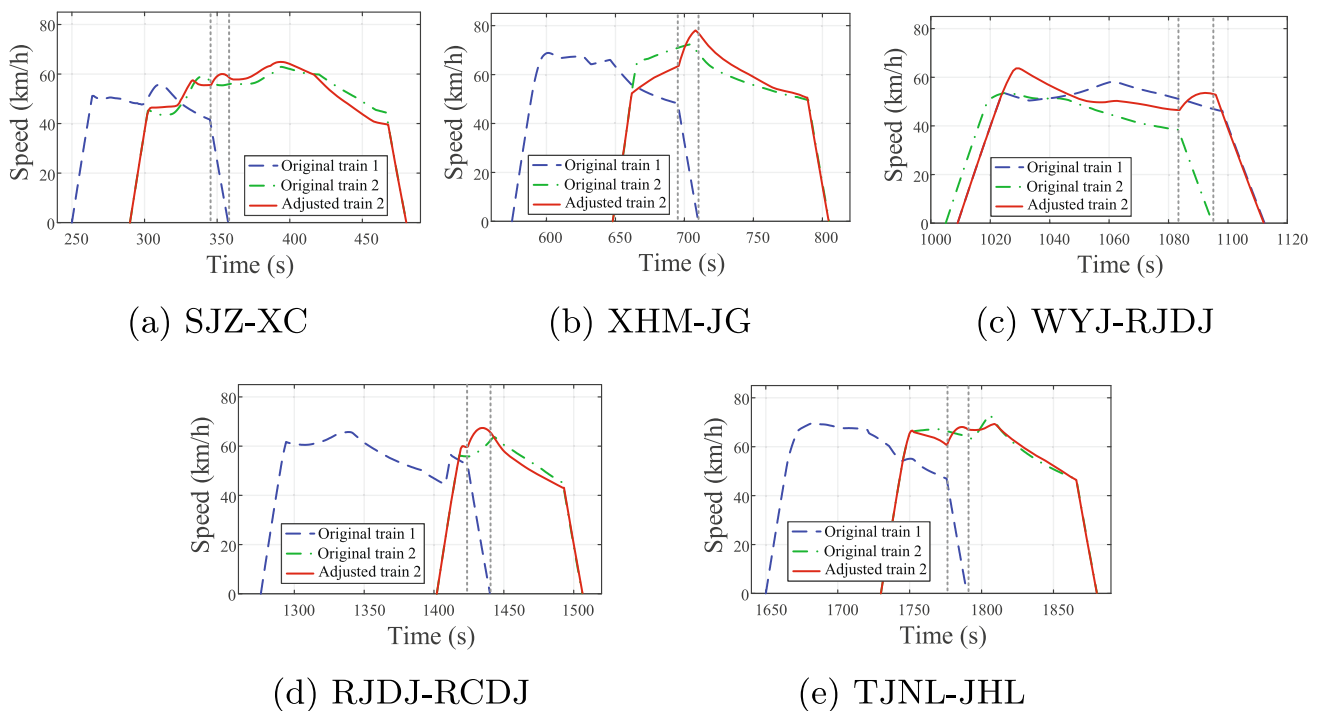
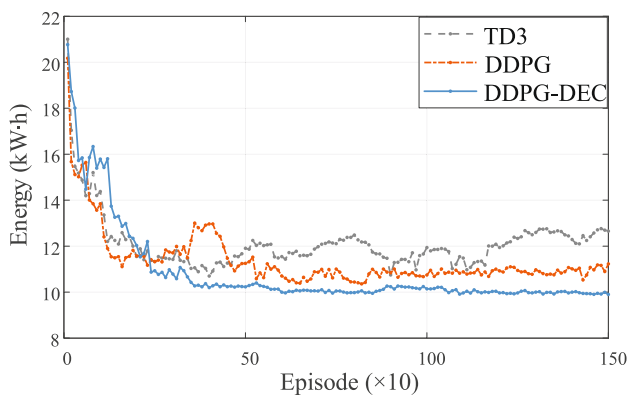
Fig. 10 Train operating speed profile and performance indicators**Fig. 11** DEC-guided train speed profile adjustment

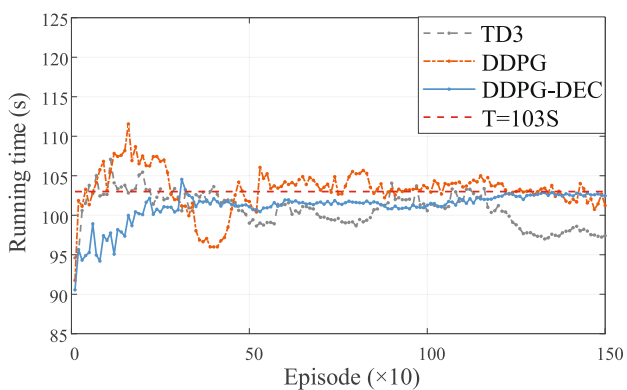
Table 2 Adjusted energy consumption comparison and energy-saving efficiency

Inter-station	$E_{\text{org}}^{\text{tr}}$ (kW·h)	$E_{\text{adj}}^{\text{tr}}$ (kW·h)	$E_{\text{net}}^{\text{tr}}$ (kW·h)	φ (%)
SJZ-XC	20.03	19.87	16.51	17.57
XHM-JG	10.39	10.05	5.50	47.06
WYJ-RJDJ	11.06	12.69	9.85	10.94
RJDJ-RCDJ	12.58	13.06	7.75	38.39
TJNL-JHL	19.07	18.87	14.63	23.28

of DEC results in a smoother and faster iterative learning process while effectively guiding the train to absorb RBE, thereby reducing net energy consumption. In Fig. 12(b), both DDPG and DDPG-DEC achieve on-time train operation, whereas TD3 exhibits significant fluctuations in travel time during training. DDPG-DEC demonstrates faster convergence, closely aligning with the scheduled travel time of 103 seconds (indicated by the red dashed line in Fig. 12(b)), leading to a notable reduction in punctuality errors. These results confirm that DEC provides stable advantages in



(a) Iterative process for energy consumption



(b) Iterative process for running time

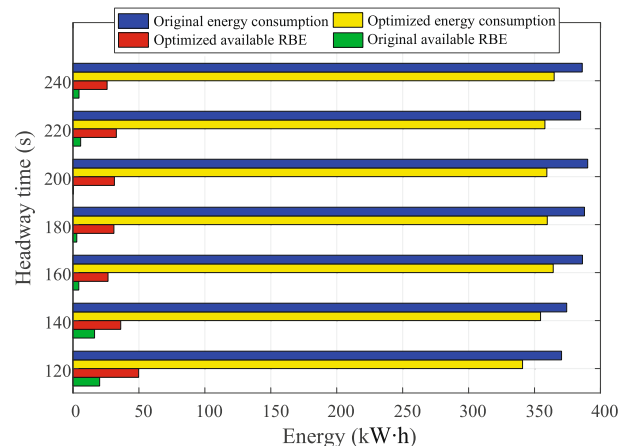
Fig. 12 Comparison of DDPG and DDPG-DEC in the WYJ-RJDJ section**Table 3** Comparison of energy consumption and t-Test results in the WYJ-RJDJ section

Algorithm	Energy (kW·h)	P-value(vs. DDPG-DEC)	Training time (h)
TD3	12.65	0.0004	0.3
DDPG	11.23	0.0085	0.3
DDPG-DEC	9.90	N/A	0.32

improving training efficiency, reducing energy consumption, and enhancing schedule adherence.

Table 3 presents the average energy consumption, corresponding t-test p-values [39], and training time for each algorithm in the WYJ-RJDJ section. In terms of energy consumption, DDPG-DEC achieves the lowest average at 9.90 kW·h, significantly lower than TD3 and DDPG. The t-test results show the p-values of 0.0004 and 0.0085 for TD3 and DDPG, respectively, both below the 0.05 significance level, confirming the statistical significance of DDPG-DEC's energy-saving advantage. Regarding training time, the algorithms show similar performance, with TD3 and DDPG requiring 0.3 hours, while DDPG-DEC is slightly higher at 0.32 hours. This suggests that the energy efficiency advantage of DDPG-DEC does not come at the expense of significant additional training time, highlighting its potential for application in cooperative control of more trains and larger-scale railway networks.

To further assess the stability and energy-saving effects of the proposed method under varying timetables, we conducted simulation experiments with seven different headway times ranging from 120 s to 240 s. We focused on two adjacent trains to analyze RBE utilization at these different intervals. Figure 13 compares the full energy composition of the two trains before and after optimization, including net energy consumption and RBE utilization. Before opti-

**Fig. 13** Comparison of energy consumption composition before and after optimization

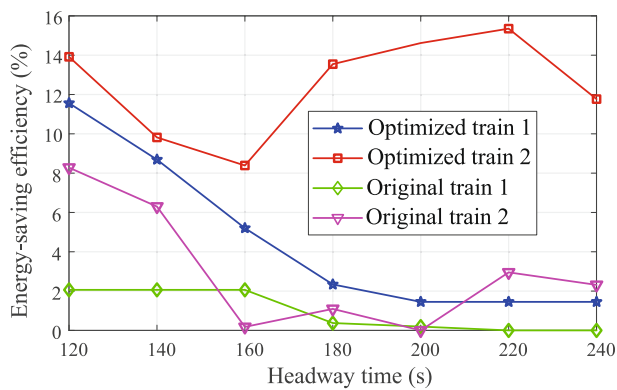


Fig. 14 Comparison of train energy-saving efficiency

mization, available RBE is generated between trains only at shorter headway times. As headway time increases, the likelihood of traction-braking overlap decreases, leading to a significant reduction in available RBE and an increase in total traction energy consumption. However, after optimizing with the multi-train cooperative control algorithm, the available RBE between trains improves considerably, resulting in a significant reduction in required traction energy consumption. Notably, even as headway time increases, the available RBE between trains remains relatively high, effectively reducing traction energy consumption.

As shown in Fig. 14, we also analyze the change of energy-saving efficiency of adjacent trains on the whole line under different headway times. When the headway time is shortened to 120 s, the trains have the best energy-saving efficiency, which can reach about 12% and 14%, respectively, after optimization. As the headway time increases, the energy-saving efficiency of optimized train 1 starts to decrease, but the energy-saving efficiency of train 2 rises and exceeds 10% after 160 s. In contrast, under the original single-train control, the energy-saving efficiency of the train does not exceed 10% at most. Comprehensively, compared with the traditional single-train control, the cooptimized train maintains a higher energy-saving efficiency on the whole and has a certain degree of robustness in the face of headway time changes.

6 Conclusions

We have proposed an energy-saving optimization model for cooperative train control, based on multi-train cooperation and heuristic rule-guided SDRL. We developed an algorithm for generating energy-efficient speed profiles that account for real-world constraints, ensuring safe, punctual, comfortable, and energy-saving subway operations. Additionally, we established a time-step-based mathematical model for RBE utilization and introduced the concept of DEC to design

the DDPG-DEC algorithm. Without altering the total train operation time, the algorithm adaptively adjusts the traction duration to maximize RBE absorption. Experimental results demonstrate that our synergetic model and algorithm can achieve energy savings of 10.94% to 47.06% between individual stations, and reduce the overall traction energy consumption of a single train by 14% or more compared to the single-train optimization approach. Future research will consider potential issues that may arise during the practical deployment of the algorithm, such as communication delays. Methods such as predictive compensation will be employed to further enhance the robustness of the algorithm in real-world applications.

Acknowledgements This work was supported in part by the National Natural Science Foundation of China under Grant 62373037, in part by the Distinguished Scientist Fellowship Program (DSFP) of Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia, and in part by Princess Nourah bint Abdulrahman University Researchers Supporting Project, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia, under Grant HCPNU2025R54.

Author Contributions Wenguang Niu: Software, Investigation, Writing - original draft preparation. Yonghua Zhou: Conceptualization, Writing - review & editing, Funding acquisition, Supervision. Xiangmeng Jiao: Investigation, Conceptualization. Hamido Fujita: Review, Funding acquisition. Hanan Aljuaid: Review, Supervision, Funding acquisition.

Data Availability Data sets during this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Informed consent was obtained from all individual participants included in the study.

References

1. Qu Y, Wang H, Wu J, Yang X, Yin H, Zhou L (2020) Robust optimization of train timetable and energy efficiency in urban rail transit: A two-stage approach. *Comput & Ind Eng* 146:106594
2. Zhou W, Huang Y, Deng L, Qin J (2023) Collaborative optimization of energy-efficient train schedule and train circulation plan for urban rail. *Energy* 263:125599
3. He D, Zhang L, Guo S, Chen Y, Shan S, Jian H (2021) Energy-efficient train trajectory optimization based on improved differential evolution algorithm and multi-particle model. *J Clean Prod* 304:127163
4. Yin J, Chen D, Li L (2014) Intelligent train operation algorithms for subway by expert system and reinforcement learning. *IEEE Trans Intell Trans Syst* 15(6):2561–2571

5. Zhang Z, Zhao H, Yao X, Xing Z, Liu X (2024) Metro timetable optimization for improving regenerative braking energy utilization efficiency. *J Clean Prod* 450:141970
6. Yang H, Li X, Yang X (2018) Mean-variance model for optimization of the timetable in urban rail transit systems. *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit* 232(4):1005–1020
7. Corman F, Meng L (2014) A review of online dynamic models and algorithms for railway traffic management. *IEEE Trans Intell Trans Syst* 16(3):1274–1284
8. Scheepmaker GM, Goverde RM, Kroon LG (2017) Review of energy-efficient train control and timetabling. *Eur J Oper Res* 257(2):355–376
9. Howlett PG, Milroy I, Pudney P (1994) Energy-efficient train control. *Control Eng Pract* 2(2):193–200
10. Liu RR, Golovitcher IM (2003) Energy-efficient operation of rail vehicles. *Transportation Research Part A: Policy and Practice* 37(10):917–932
11. Albrecht A, Howlett P, Pudney P, Vu X, Zhou P (2016) The key principles of optimal train control-part 2: Existence of an optimal strategy, the local energy minimization principle, uniqueness, computational techniques. *Transportation Research Part B: Methodological* 94:509–538
12. Lechelle S, Mouneimne Z (2010) Optidrive: A practical approach for the calculation of energy-optimised operating speed profiles. In: *Iet conference on railway traction systems (rts 2010)*, pp 1–8. IET
13. Lu S, Hillmansen S, Ho TK, Roberts C (2013) Single-train trajectory optimization. *IEEE Trans Intell Trans Syst* 14(2):743–750
14. Sicre C, Cucala A, Fernández-Cardador A (2014) Real time regulation of efficient driving of high speed trains based on a genetic algorithm and a fuzzy model of manual driving. *Eng Appl Artif Intell* 29:79–92
15. Fang J, Guo X, Liu Y, Chang X, Fujita H, Wu J (2023) An attention-based deep learning model for multi-horizon time series forecasting by considering periodic characteristic. *Comput & Ind Eng* 185:109667
16. Huang C, Li M, Cao F, Fujita H, Li Z, Wu X (2022) Are graph convolutional networks with random weights feasible? *IEEE Trans Pattern Anal Mach Intell* 45(3):2751–2768
17. Zhou K, Song S, Xue A, You K, Wu H (2020) Smart train operation algorithms based on expert knowledge and reinforcement learning. *IEEE Trans Syst, Man, and Cybern: Syst* 52(2):716–727
18. Shang M, Zhou Y, Fujita H (2021) Deep reinforcement learning with reference system to handle constraints for energy-efficient train control. *Inf Sci* 570:708–721
19. Zhang L, Zhou M, Li Z et al (2021) An intelligent train operation method based on event-driven deep reinforcement learning. *IEEE Trans Ind Inf* 18(10):6973–6980
20. Lin X, Liang Z, Shen L, Zhao F, Liu X, Sun P, Cao T (2023) Reinforcement learning method for the multi-objective speed trajectory optimization of a freight train. *Control Eng Pract* 138:105605
21. Gu S, Yang L, Du Y, Chen G, Walter F, Wang J, Knoll A (2024) A review of safe reinforcement learning: Methods, theories and applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*
22. Wang D, Wu J, Wei Y, Chang X, Yin H (2024) Energy-saving operation in urban rail transit: A deep reinforcement learning approach with speed optimization. *Travel Behav Soc* 36:100796
23. Zhao Z, Xun J, Wen X, Chen J (2022) Safe reinforcement learning for single train trajectory optimization via shield sarsa. *IEEE Trans Intell Trans Syst* 24(1):412–428
24. Liu Y, Halev A, Liu X (2021) Policy learning with constraints in model-free reinforcement learning: A survey. In: *The 30th international joint conference on artificial intelligence (ijcai)*
25. Achiam J, Held D, Tamar A, Abbeel P (2017) Constrained policy optimization. In: *International conference on machine learning*, pp 22–31. PMLR
26. Yang L, Ji J, Dai J, Zhang L, Zhou B, Li P, Yang Y, Pan G (2022) Constrained update projection approach to safe policy optimization. *Adv Neural Inf Process Syst* 35:9111–9124
27. Sun P, Zhang C, Jin B, Wang Q, Geng H (2023) Timetable optimization for maximization of regenerative braking energy utilization in traction network of urban rail transit. *Comput & Ind Eng* 183:109448
28. Sahebi H, Ashenaei H, Mirzamohammadi S, Gilani H (2023) Reducing passenger waiting time and energy consumption for metro timetabling: A robust optimization approach. *Energy Reports* 10:4051–4062
29. Shang M, Zhou Y, Fujita H (2021) Energy-saving operation synergy for multiple metro-trains using map-reduce parallel optimization. *IEEE Trans Veh Technol* 71(2):1319–1332
30. Liu J, Guo H, Yu Y (2016) Research on the cooperative train control strategy to reduce energy consumption. *IEEE Trans Intell Trans Syst* 18(5):1134–1142
31. Sun X, Cai H, Hou X, Zhang M, Dong H (2014) Regenerative braking energy utilization by multi train cooperation. In: *17th International IEEE conference on intelligent transportation systems (ITSC)*, pp 139–144. IEEE
32. Jin B, Fang Q, Wang Q, Sun P, Feng X (2021) Energy-efficient train control in urban rail transit: multi-train dynamic cooperation based on train-to-train communication. In: *2021 IEEE intelligent vehicles symposium (IV)*, pp 309–314. IEEE
33. Su S, Wang X, Tang T, Wang G, Cao Y (2021) Energy-efficient operation by cooperative control among trains: A multi-agent reinforcement learning approach. *Control Eng Pract* 116:104901
34. Hayashi T, Fujita H, Hernandez-Matamoros A (2021) Less complexity one-class classification approach using construction error of convolutional image transformation network. *Inf Sci* 560:217–234
35. Lillicrap T.P, Hunt JJ, Pritzel A, Heess N, Erez T, Tassa Y, Silver D, Wierstra D (2015) Continuous control with deep reinforcement learning. [arXiv:1509.02971](https://arxiv.org/abs/1509.02971)
36. Yang X, Li X, Ning B, Tang T (2015) An optimisation method for train scheduling with minimum energy consumption and travel time in metro rail systems. *Transportmetrica B: Transport Dynamics* 3(2):79–98
37. Liao L, Zeng J, Lai T, Xiao Z, Zou F, Fujita H (2024) Stereo matching on images based on volume fusion and disparity space attention. *Eng Appl Artif Intell* 136:108902
38. Fujimoto S, Hoof H, Meger D (2018) Addressing function approximation error in actor-critic methods. In: *International conference on machine learning*, pp 1587–1596. PMLR
39. Berrar D (2022) Using p-values for the comparison of classifiers: pitfalls and alternatives. *Data Min Knowl Disc* 36(3):1102–1139

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Wenguang Niu received the B.S. degree in Automation in 2023 from the Qingdao University, Qingdao, China, where he is currently pursuing the master's degree from Beijing Jiaotong University, Beijing, China. His research interests include deep reinforcement learning, intelligent control and train energy-saving optimization.



Yonghua Zhou (Member, IEEE) received the B.S. degree in automatic control from Beijing University of Aeronautics and Astronautics, China, in 1991, and the Ph.D. degree in control theory and control engineering from Tsinghua University, China, in 2003.

From 1996 to 1997, he studied at Tokyo Denki University, Tokyo, Japan. From 2003 to 2006, he was a Postdoctoral Researcher with the Institute of Transportation Engineering, Tsinghua University. From 2011 to 2012, he was a Visiting Scholar with the Center for Systems Science, Yale University. He is currently a Professor of traffic information engineering control with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing, China. He has published more than 60 papers in the international journals and conferences. His research interests include artificial intelligence, parallel computing, cloud computing, optimization control theory, and intelligent transportation systems.



Xiangmeng Jiao received the B.S. degree in 2017 in rail traffic signaling and control from the Henan Polytechnic University, Henan, China, and the M.S. degree in 2020 in traffic information engineering and control with Automatic Control Institute from the Lanzhou Jiaotong University, Gansu, China. He worked as a railway signal design engineer in 2021 at CASCO Signal Co., Ltd. in Beijing, China. He is currently pursuing the Ph.D. degree in traffic information engineering and control with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing, China. His research interests include energy-efficient train control, energy-efficient train timetables, integrated optimization, artificial intelligence, machine learning, optimization algorithms, reinforcement learning, parallel computing, and fault diagnosis.

neering and control with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing, China. His research interests include energy-efficient train control, energy-efficient train timetables, integrated optimization, artificial intelligence, machine learning, optimization algorithms, reinforcement learning, parallel computing, and fault diagnosis.



Hamido Fujita (Life Senior Member, IEEE) received the B.S. degree in electrical engineering from the University of Manchester, Manchester, U.K., in 1979, and the master's and Ph.D. degrees in information engineering from Tohoku University, Sendai, Japan, in 1985 and 1988, respectively, and the Doctor Honoris Causa from Óbuda University, Budapest, Hungary, in 2013, and Timisoara Technical University, Timisoara, Romania, in 2018.

He is currently a Professor of artificial intelligence with Iwate Prefectural University, Takizawa, Japan, where he is the Director of intelligent software systems. He is also an Adjunct Professor of computer science and artificial intelligence with Stockholm University, Stockholm, Sweden; the University of Technology Sydney, Ultimo, NSW, Australia; National Taiwan Ocean University, Keelung City, Taiwan; and the University of Granada, Granada, Spain. He has supervised Ph.D. students jointly with Laval University, Québec City, QC, Canada; the University of Technology Sydney; Oregon State University, Corvallis, OR, USA; the University of Paris 1 Pantheon-Sorbonne, Paris, France; and the University of Genoa, Genoa, Italy. He has given many keynotes at many prestigious international conferences on intelligent systems and subjective intelligence. He headed a number of projects, including intelligent HCI, a project related to mental cloning for healthcare systems as an intelligent user interface between human users and computers, and a SCOPE project on virtual doctor systems for medical applications. He has four international patents in software systems and several research projects with Japanese industry and partners.

Dr. Fujita was a recipient of title of the Honorary Professor from Óbuda University in 2011 and the Honorary Scholar Award from the University of Technology Sydney in 2012. He is also a Highly Cited Researcher in cross-field in 2019 and computer science in 2020, 2021, 2022, 2023, and 2024 by Clarivate Analytics. He is the Chairperson of the i-SOMET Incorporated Association. He was the Editor-in-Chief of *Knowledge-Based Systems* from 2002 to 2019 and he is the Emeritus Editor of *Knowledge-Based Systems*. He is also the Editor-in-Chief of *Applied Intelligence* (Springer). He is also the Vice President of the International Society of Applied Intelligence.

Hanan Aljuaid received the B.S. degree from KAU University and the M.S. and Ph.D. degrees in computer science from UTM University, in 2014. She is currently with the Computer Sciences Department, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University (PNU), Saudi Arabia. She has published numerous articles on pattern recognition, the IoT, and data science. Her research interests include computer vision and NLP. Much of her work has been on improving the understanding, design, and performance of pattern recognition, mainly through the application of data mining and machine learning. She has given numerous invited talks and tutorials.